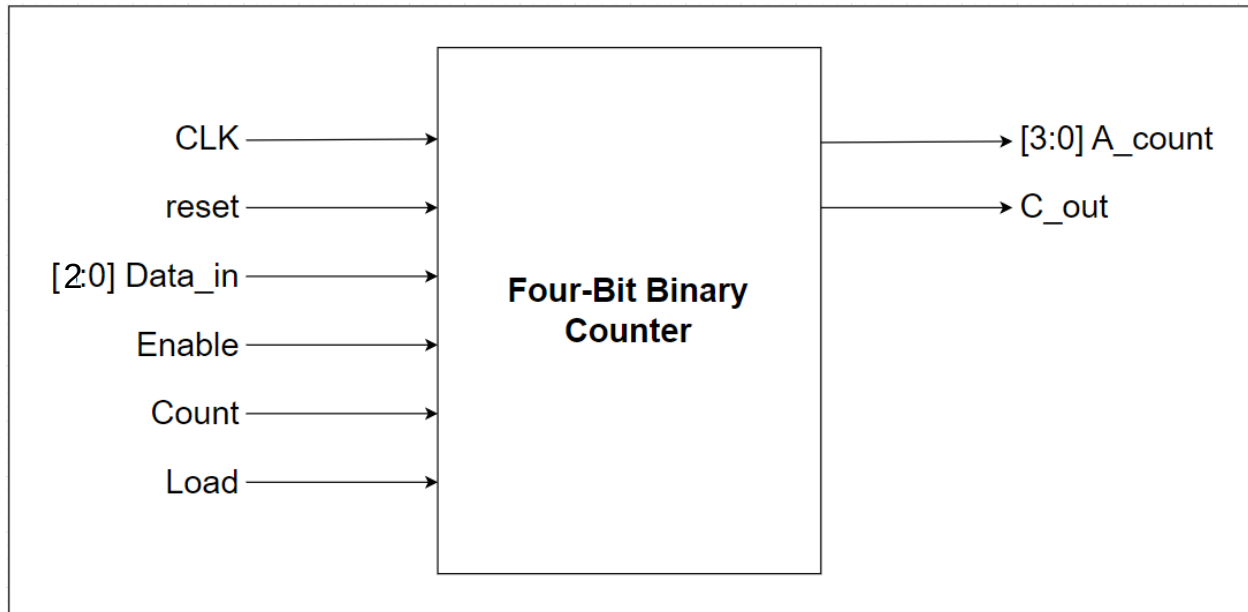


4-Bit Binary Counter Specifications

Top Module:



The Top Module of a Four-bit binary counter is given above.

Functionalities:

1. Synchronous reset
2. Load data into the register
3. Up-counting operation
4. Down-counting operation

Signal List:

Signals	Bit Width	Description
CLK	1	Input Clock Frequency 50~100MHz.
reset	1	Input control signal. Active Low synchronous reset signal which resets the circuit when reset = 0.
Enable	1	Input control signal. It works as Chip Enable. Enable = 0, The circuit keeps its current state of A_count and C_out. Enable = 1, data loading, up-counting, and down-counting operations will work.
Load	1	Input control signal. It is used to load data_in in the circuit. Load = 0, The circuit keeps its current state. Load = 1, then load Data_in data into the circuit.
Data_in	3	A 3-bit input data signal.
Count	1	Input control signal. Controls the up-counting/down-counting. Count = 0, down-counts the current data, -1 Count = 1, up-counts the current data, +1
A_count	4	A 4-bit output data signal.
C_out	1	An output data signal. C_out is an output carry signal that indicates whether the counter (A_count) has exceeded its maximum value. C_out = 1: A_count has surpassed its maximum value. C_out = 0: A_count is yet to reach its maximum value.